

1. Find the page fault ratio and Hit ratio using FIFO, LRU, MRU and optimal algorithms with memory size = 3 and Sequence is 3 2 1 3 3 1 6 2 4 3 4 2 1 4 5 2 1 3 4

A:- Number of frames = 3

Total page references = 19

$$\text{Hit Ratio} = \frac{\text{Hit}}{19}$$

$$\text{Fault Ratio} = \frac{\text{Fault}}{19}$$

1. FIFO

	3	2	1	3	3	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	4	4	6	6	4	2	4	4	4	4	5	5	5	3	4
	X	X	✓	X	✓	X	✓	X	X	X	✓	✓	✓	X	✓	✓	X	X	X

Hit = 8

Faults = 11

$$\text{Hit Ratio} = \frac{8}{19}$$

$$\text{Fault Ratio} = \frac{11}{19}$$

2. LRU

	3	2	1	3	3	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	1	1	4	4	4	4	4	4	4	4	1	1	1
2	-	2	2	2	4	4	4	2	2	2	2	2	2	2	5	5	5	3	3
3	3	3	3	3	3	3	6	6	6	3	3	3	1	1	1	2	2	2	4
	X	X	X	✓	X	✓	X	X	X	X	✓	✓	X	✓	X	X	X	X	X

Hit = 5 Faults = 14

$$\text{Fault Ratio} = \frac{14}{19}$$

$$\text{Hit Ratio} = \frac{5}{19}$$

3. MRU

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6
2	-	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	3	4
3	3	3	3	3	4	4	4	4	4	3	4	4	4	4	5	2	2	2	2
	X	X	X	✓	X	✓	X	✓	✓	X	X	✓	X	✓	X	X	✓	✗	✗

Hits = 7

Page faults = 12

Hit Ratio = $7/12$

Page Fault Ratio = $12/19$

4. Optimal Page Replacement :-

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	1
2	-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	4	4	4	4	4	4	4	4	4	4	5	5	5	3	4
	X	X	X	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	X	✓	✓	X	X

Hits = 9

Faults = 10

Hit Ratio = $9/19$

Fault Ratio = $10/19$

Efficient $\rightarrow$ Optimal.

Apply FIFO, LRU and MRU and check whether Belady's anomaly exists or not for physical Memory 3 frames and 4 frames

1 2 3 4 1 2 5 1 2 3 4 5

A:- Total page references = 12

Number of frames = 3

	1	2	3	4	1	2	5	1	2	3	4	5
F ₃	-	-	3	3	3	3	3	3	3	3	3	3
F ₂	-	2	2	2	2	2	2	2	2	2	2	2
F ₁	1	1	1	4	1	1	5	1	1	1	4	5
	x	x	x	x	x	x	x	x	✓	x	x	x

Faults = 9

Hits = 3

Hit Ratio = $3/12$

Fault Ratio = $9/12$

For 4 Frames

	1	2	3	4	1	2	5	1	2	3	4	5
F ₄	-	-	-	4	4	4	5	4	4	3	3	3
F ₃	-	-	3	3	3	3	3	3	2	2	2	2
F ₂	-	2	2	2	2	2	2	2	1	1	1	5
F ₁	1	1	1	1	1	1	5	5	5	5	4	4
	x	x	x	x	✓	✓	x	x	x	x	x	x

Faults = 10

Hits = 2

Hit Ratio = $2/12$

Fault Ratio = $10/12$

Belady's Anomaly exists for FIFO because

3 frames → 9 faults

4 frames → 10 faults,

So, increasing frames increases faults

LRU for 3 frames

	1	2	3	4	1	2	5	1	2	3	4	5
F ₃	-	-	3	3	3	2	2	2	2	2	2	2
F ₂	-	2	2	2	1	1	1	1	1	1	4	4
F ₁	1	1	1	4	4	4	5	5	5	3	3	2
	x	x	x	x	x	x	x	✓	✓	x	x	x

Faults = 10

Hit = 2

Hit Ratio = $2/12$

Fault Ratio = $10/12$

MRU

	1	2	3	4	1	2	5	1	2	3	4	5
	-	-	3	4	4	4	4	4	4	4	4	4
	-	2	2	2	2	2	5	5	5	5	5	5
	1	1	1	1	1	1	1	1	2	3	3	3
	x	x	x	x	✓	✓	x	✓	x	x	✓	✓

Faults = 7

Hit = 3

Fault Ratio = $7/12$

Hit Ratio = $3/12$

Most Efficient → MRU